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An Automated Student Attendance Tracking System Based on Voiceprint and Location

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***Abstract*—Taking attendance has been widely used as a method to track students' academic behaviours. But conventional approaches are usually time consuming and inaccurate. In this paper, we have proposed an automated attendance tracking system, where students can use smartphones to submit attendance in parallel. Identity of a student is collaboratively verified by voiceprint and realtime location. Service isolation and token-based mechanism is applied specifically in the system as well, for the purpose of protecting users' privacy and the system's security. Tests in real world environment has demonstrated that our system is characterised as high efficiency and accuracy in tracking attendance.**

***Index Terms*—attendance tracking; biometrics; voiceprint; location-based services**

I. Introduction

Student involvement plays an important role in the success of a learning process [1]. And attendance in a class has been used for a long time to measure students' involvement in most high schools and universities. Gump [2] has highlighted a strong relevance between students' class attendance and academic performance that poor performance is usually connects with poor attendance. Besides, being absent in classes not only can affect individual's schoolwork, but also can hurt the overall academic environment. Thus, reducing the rate of absence has been an essential but challenging work for every school.

Traditionally, tracking attendance in a class is accomplished manually by lecturer, passing an attendance sheet around the classroom during lecture, and asking students to sign their names. Disadvantages of this method has been revealed in practice. For instance, such method cannot prevent forgery of registering fake attendance for someone who do not appear in the classroom. Moreover, passing sheet would distract students from lecture content, and in some extent it would affect the quality of the class. Another intuitive way is roll-call. The lecturer can mark attendance by calling names of each students one after another. This method can obtain accurate attendance records, while it is unacceptable in classes containing hundreds of students, which could even consuming half of the time in a lecture. Apart from these issues, processing attendance records manually after lecture is time-consuming. Teachers would put much effort on calculating and integrating attendance records with other academic measurement indicators. As a result, developing automated attendance systems would make this process efficient and help improve the quality of the class.

There were many studies aiming at building automated attendance tracking systems, and many of them have been implemented in the real world environment successfully. Normally, approaches solving this problem could be classified into RFID/NFC-based and visual-based methods. RFID technology was widely used in many scenes in our daily life from apartment security tags to payment card [3]. Students using RFID/NFC-based systems would be equipped with a RFID tag containing unique registration code, or directly write the code into their cellphones using NFC technology. By hovering tags or cellphones upon the RFID reader placed in every classroom, students' appearance in a class would be recorded instantly. Silva et al. [4] and Chang [5] introduces the method of applying RFID technology into classes. They have utilised RFID readers connected to the Internet, and attendance records would be transmitted to a backend server. Zhi et al. [6] on the other hand employed NFC modules inside modern cellphones integrated with secret code for authentication. MB & Roy [7] provided a web-based platform collecting and analysing records in real time, along with SMS and email notification functionalities, which was distributed into real environment and indicated its efficiency. However, using RFID technologies involves installations of specific readers in the classroom, which leads to expenses on hardware and engineering works. Besides, these methods also have less considerations of student with multiple RFID cards or card losing situation, which cannot accurately authorised student's identity.

Visual-based systems usually utilise cameras installed in the classroom to capture photos, then face recognition algorithms could be used to identify face biometrics determining each student's attendance. Oktavianto et al. [8] proposed a real time video analysis system for logging attendance, which could run at 16 frames per second. Instead of using single camera, Park et al. [9] has implement a system using three Kinect to track students' appearance, providing much higher accuracy in real environment test. Nevertheless, face recognition algorithms are far from perfect at present, and engineers have to deal with many issues. Bad light conditions, different facing angles and poses the student might cause a seriously decrease in accuracy of the result. In real world applications, such inaccuracy means even more manual work to check and correct the records.

In this work, we propose to utilise both voiceprint technology and location-based services collaboratively for tracking students' attendance in a class. Lecturers and students use this

system through applications on their smartphones. Backend servers are responsible for gathering and storing records, which could be accessed by lecturers through a web-based portal in both raw and interactively statistical way.

The rest of the paper is organised as follows. Section II explains the key problems associated with the current implementation of attendance tracking systems, which motivates our work. Section III introduces the architecture of our proposed system. Section IV discusses about the detailed implementations from many aspects. Section V presents the result and applications in real world environment. Finally, section VI summarises the paper and discusses our future plan.

II. Motivation

Current students' attendance tracking methods and existing automated systems are somewhat having drawbacks. We have identified and analysed several key factors dominating attendance tracking procedure in this section, which are listed below:

Identity The most important factor of taking attendance is to verity the identity of a student. A student may intentionally marking attendance for their missing classmates, which in most cases remain unnoticed by the lecturer.

Location Any Internet-based work can be completed remotely. A student who passed identity authentication may not stay in the classroom. Ensuring students who submit the attendance records are actually in the classroom is rather important.

Data management Not only the operations of taking attendance itself, processing, analysing and storing data are all challenging works. Lecturers may take hours to calculate in order to get an overall view of attendance condition in the class. At the end of the semester, making those records reflect on students' grade is time-consuming as well.

The goal of work is to make students submitting their attendances by using the most commonly used smartphones. To achieve identity authentication, we have chosen the voiceprint verification technology. Voice biometrics, like fingerprint, has unique characteristics which could be used to identify a person. Using voiceprint as an indicator of user's identity have been widely used in many occasions such as access control and banking scenarios, which enhances security as well as promotes efficiency. So that we can use voiceprint verification to verify students' identities during attendance tracking procedure.

In terms of location, location-based services has been widely utilised in mobile applications for years. With the help of them, precise location of the cellphone could be obtained when a student is about to submit his attendance. Collaboratively using voiceprint and realtime location as a verification method could make attendance tracking more efficient and reliable. Firstly, both voiceprint and realtime locations could be easily collected in ordinary smartphones, so that there is no need for any tags or devices carried by students or installed in classrooms. Secondly, such method enables attendance taking to be operated in parallel. Students could complete all the verifying and recording steps on their smartphones, instead of standing in lines to use the specific RFID readers in conventional systems. Using less time in attendance tracking provides teachers with more flexibility and opportunity to conduct the lecture, which leads to higher student involvement in the class.

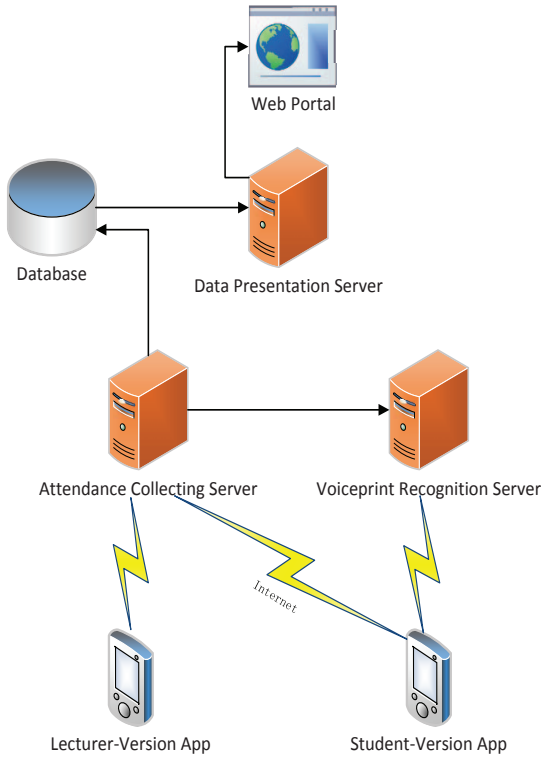

Fig. 1. Architecture of the system

III. System Architecture

Our proposed system is made up of several connected components. Firstly, smartphone applications is used for both students marking attendance and lecturers conducting the process. In addition, an online voiceprint verification service would be established for students' voiceprint verification. At last, distributed servers are used for collecting, storing and analysing attendance records submitted from students' smartphones.

The architecture of our proposed system is shown in Fig. 1. The whole system has been designed in a client-server pattern. The client side contains the applications in smartphones, where we have separated them into student-version and lecturer-version for different purposes. The other side of the system is made up of a voiceprint recognition server, an attendance collecting server, a database server and a data presentation server. The clients and servers is connected by the Internet connection, relying on cellphones' cellular or WLAN network.

The smartphone applications are designed for both students and lecturers accessing the system, in order to complete attendance tracking. Two versions of applications have been developed for different purposes. The student-version application is responsible for capturing students' voice and submitting

verified attendance records. On the other hand, lecturer-version application is usually used for controlling lifecycle of one checking period.

The voiceprint recognition server is responsible for verifying students' voiceprint. Once a student wants to submit his presence in the class, the voiceprint would be used to verify his identity. After the server receives a recognition request, the captured voice clip and student's stored voiceprint template will be compared to determine if they are come from the same person. If it is, the message needed for further attendance submission is returned to the client, otherwise the server will return a verification failure message.

The attendance collecting server receives the presence submission as well as it controls the primary logic of the attendance tracking process. Lecture-version app communicates with the server controlling beginning and ending of the activity, while the student-version app fetches the courses list from the server, and submit verified attendance records to it. All the received attendance records will be sent to the database.

The database server stores all the attendance records. Each record contains a number of necessary fields, such as student-id, course-id and time of the submission. Furthermore, course related information including lecture time, location, and students' list of this course is also stored in the database, which links the attendance records with the course, and enables the lecturer to access the data intuitively.

The data presentation server provides all functionalities for accessing the attendance data. The server reveals its functionalities in the from of a web-based portal. By communicating with the database, the server obtains courses info and students' attendance records, then renders them in the web portal. In addition to presenting raw records, the portal has several built-in statistical methods, and results are displayed through interactive plots on the page. Finally, attendance data could be exported and downloaded in the portal as well.

IV. Implementation

A. Initial data registration

Before utilising our system, both students and lecturers need to register necessary information. For a student, registration is mainly about personal information. Basic data such as student-id and department would be stored into the database. The student also needs to walk through a voiceprint registration process in which a voiceprint template will be collected and saved into the voice recognition service. All registration processes are encouraged to be done in students' own smartphones, for eliminating the deviation caused by different microphones. In this manner, the registration process could perform efficiently in parallel, without spending lots of time organising specific events.

On the lecturer's side, the registration is concentrated on course information. In order to use the system, a lecturer should firstly add the course information into the system, including course name, location, time and students' list. Students' list makes it convenient for lecturers to browse attendance records, as well as prevent students submitting presence

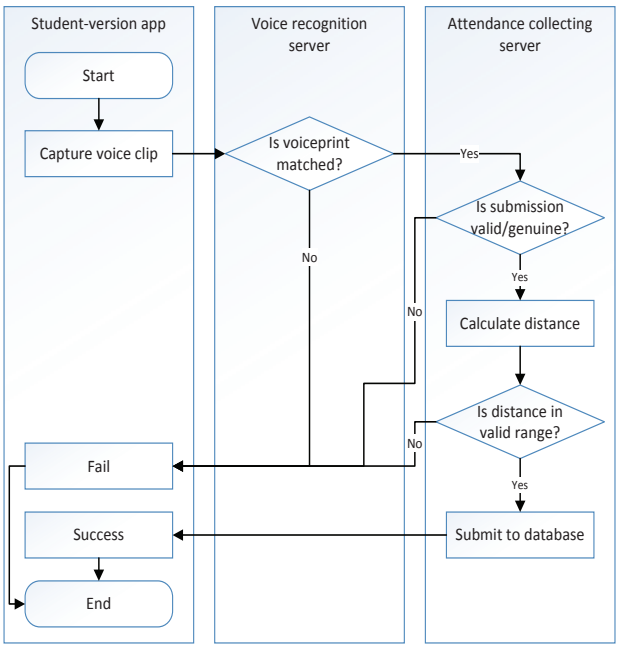

Fig. 2. The process of students attendance submission

to classes that do not enrol. Different from the students' registration, lecturers could register the course through web portal from the personal computer.

Note that we have separated students' voiceprint template apart from the courses data in our system. The voiceprint represents the students' identity, and identity is independent from a specific course. So that it is possible for students to register only one template supporting verification for all the courses, which considerably reduces the complexity of the system.

B. Workflow of the attendance recording

Students and lecturers could use our automatic attendance tracking system efficiently. The workflow of the attendance recording procedure is shown in Fig. 2. Lecturers could control the switch that determines the status of a attendance checking procedure. And the switch can be modified from the lecturer-version app through a web service provided by attendance collecting server. Cellphone's location of the lecturer will also be sent to the server, and such data would be marked as the location of the classroom.

Students use the application in their smartphones to submit presence in the class. When a student enters the application and selects the course, his voiceprint would firstly be captured and sent to the voice recognition server for verification. If it is passed, the app would form a message containing student's info with cellphone's current location, and send it to the attendance collecting server. After receiving the message, the server needs to ensure it to be valid and genuine, which could protect the accuracy and reliability of the records in the system. Detailed security control implementations would be discussed in Section D. Then the server would calculate and check the distance between student and course location. Due to the

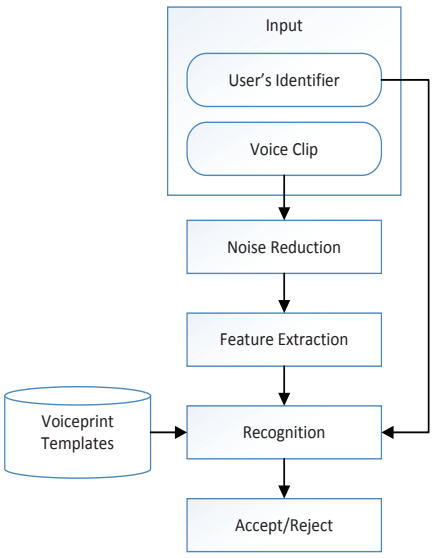

Fig. 3. The workflow of voiceprint verification

deviation of positioning services, we have setup a range of tolerance in location. If the distance check is successful, the attendance records would be finally submitted to the database.

C. Voiceprint and location verification

There is a trade-off between promoting efficiency and ensuring result's accuracy in attendance tracking scenario. We have proposed to integrate the use of voiceprint biometrics with realtime location for keeping attendance records accurate. Voiceprint indicates the identity of the student, while location reflects whether the student is sitting in the classroom. Collaboratively verification by combining these two technologies in one workflow could in some extent minimise the possibility of error and forgery in tracking attendance.

The voiceprint verification service is an independent component of the system. Since biometrics like voiceprint are deeply connected with users' privacy and can be reused in many sensitive occasions [10], protecting voiceprint related data and algorithms are highly important in system design. By exposing only crucial interfaces, we have isolated the voice verification service with other system components for security. The workflow of a typical voiceprint verification procedure has been shown in Fig. 3. The captured voice clip would be processed for noise reduction at first. And feature extraction produces formatted data which simplifies the recognition process. At last, template marked as the caller's identifier will be compared with the formatted data, generating a match score. If the score is greater than the pre-defined threshold, then the service could assume that the person who request the service actually owns that account. We have developed our voice recognition services relying on speaker recognition API of Microsoft cognitive services [11].

In terms of the location verification, we have employed location-based services provided by Baidu Inc. Positioning SDK is used for obtaining location of the cellphone. Normally the GPS signal is too weak to get accurate location data in the classroom. However, a creative method of integrating results from GPS, cellular network, and Wi-Fi signals has been implemented by Positioning SDK, so that we can utilise this ability to accurately track where the students are. On the server side, the system would determine whether the distance between two locations' is in the tolerable range. Geography systems usually have complex methods and algorithms to calculate distance between two coordinates. Relying on Web Services API, we could access the geographic functionalities from our server through HTTP interfaces exposed by Baidu cloud services.

D. Token-based security

The separation of voiceprint verification from attendance records submission has largely reduced the systems' complexity as we mentioned above. However, a security concern might be raised by this approach. Since two services is distributed on different machines, network requests should be made for identity verification and record submission respectively. Taking the second request into account, it is not safe when someone has the ability of generating fake submission requests. Manually making fake requests could bypass the voiceprint verification mechanism and submit attendance, which is a huge drawback of design in the system. Zhi et al. [6] has proposed the idea of using dynamically generated secret code to prevent forgery. Hence we introduce the token-based mechanism to promote the reliability of the submission.

In our implementation, the voiceprint recognition server establishes an authenticated connection with the attendance collecting server. Once the voiceprint recognition server successfully verified a voiceprint, a dynamically generated token would be retrieved from the attendance collecting server, and returned to the client app. Then client would combine the token with submission message and send them to the attendance collecting server. After receiving a submission request, token would firstly be verified with the stored one in order to authenticating whether the request is genuine. If it is, the subsequent verification and submission could be operated.

Every successful voiceprint verification will retrieve an one-time available token, which requires tokens to be unique during each generation. To produce a distinct token, a string is created by concatenating basic information of the student, such as student-id, name, and department. Then a randomly generated salt is added to the end of the string. At last, a 32-character-long token could be generated by feeding the string into a MD5 function.

E. Web-based portal

Lecturers are able to access the data of students attendance through a web-based portal. They can query attendance records of a particular student either by his name or student-id. Or the lecturer can just list out the attendance percentage of every student in the class, which is automatically calculated. All the data could be viewed on month or semester basis. Besides listing raw data, several statistical result would be calculated and presented in interactive plots along with the list.

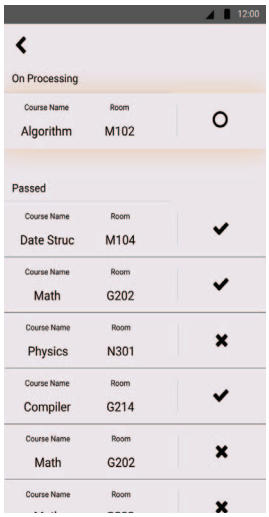

Fig. 4. Course selection

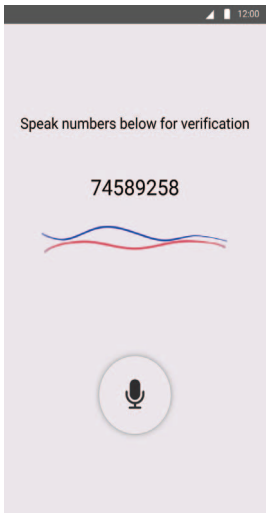

Fig. 5. Voiceprint verification

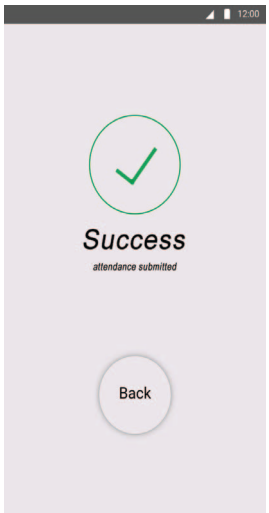

Fig. 6. Successful submission

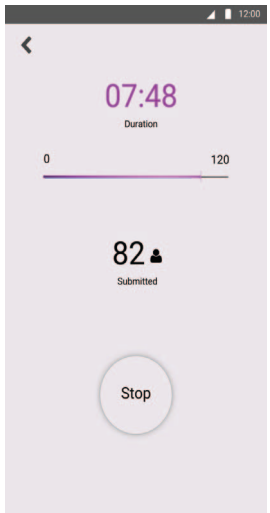

Fig. 7. Process of checking activity

In the implementation, we have designed the portal to be a single-page application based on React and Flux frameworks, which are open sourced by Facebook. The portal is hosted on the data presentation server, while the server also provides web services for querying data from the database.

V. Result and Test

In this section, our implemented system will be introduced. Then the system would be applied into a real world environment to measure its efficiency in students attendance tracking operations.

We have developed the smartphone applications based on Android platform. The screenshots of our implemented attendance tracking system is shown in figures below. Fig. 4 - 7 illustrates the user interfaces of smartphone applications. The course which is available for attendance tracking is highlighted on screen, while the past attendance checking activities and status is listed below, as shown in Fig. 4. After picking the course to submit attendance, students will follow the instructions shown in Fig. 5 for verifying voiceprint. Then the successive submission and verification processes will be controlled by the app. Once the submission could be verified and stored in the database, a successful message will be displayed on screen like Fig. 6, indicating the completion of the process. Apart from others, Fig. 7 shows the UI presented during attendance tacking in the lecturer-version application. Lecturers could monitor the number of students who have already submitted attendance, as well as halt the activity by pressing the stop button.

Fig. 8 shows the screenshot of the web-based portal in a browser. Layout of the page could be divided into three parts vertically. The upper part lies a form for lecturers specifying data query conditions. The lecturer could choose to query data of a specific student, or just list all the records of a class. The middle of the page is responsible for displaying the queried data, where raw records are listed on the left, and automatically generated statistical plots are presented on the right. At the bottom of the page, the export button provides lecturers with an interface for exporting records and downloading them in the form of spreadsheets.

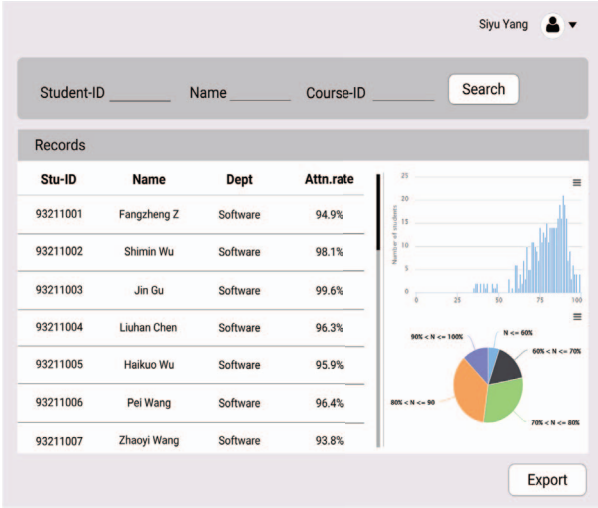

Fig. 8. Screenshot of the web-based portal

To examine the efficiency of the system, we have applied it into a real world environment. We focused our test in an undergraduate course of computer science with around 120 students. The test has lasted for weeks. Statistics gained from the test has proved several characteristics of our proposed system. Firstly, we have succeeded in providing precise attendance records in most occasions, while there are some cases that external conditions such as cellphones' power and network may affect the workflow we have designed. Secondly, time consumed in every attendance tracking procedure has considerably reduced compared with conventional approaches. In average, it takes less than 5 minutes to finish a process in our test. Moreover, we have noticed a phenomenon that the longer system is used, the less time it costs for every tracking procedure. The main reason behind this is the familiarity that students can execute faster when they get familiar with the

procedure. It turns out that our system may perform much better if widely being used in different classes.

VI. Conclusion

Tracking students' attendance in a class has always been a complex and time consuming work. In this work, we have designed and implemented an automated attendance tracking system. Students complete the attendance taking procedure on their smartphones in parallel. Collaboratively verification by integrating voiceprint biometrics with realtime location could maximise the accuracy of the result collected by the system, where several methods have been applied to protect privacy-sensitive biometrics data. In addition, a web-based portal enable lecturers to access the records in a more intuitive and interactive way. Results have shown the high efficiency and usability of our system in real world environment. The methods and system we proposed in this paper can be further improved in many aspects. Future work of the attendance tracking system will mainly focus on data processing and analysis techniques, which aims at providing lecturers and universities with intelligent assistance in monitoring students' global academic behaviour.

Acknowledgment

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